



Conditional and Control Statements

Topics:

- Understanding conditional and control statements
- Examples

Conditional Statement

What are Conditional Statements in Python?



Conditional Statement in Python perform different computations or actions depending on whether a specific Boolean constraint evaluates to true or false. Conditional statements are handled by IF statements in Python.

Conditional Statement

Python if Statement Syntax:

```
if expression:  
    Statement  
else:  
    Statement
```

Indentation

Python relies on indentation (whitespace at the beginning of a line) to define scope in the code. Other programming languages often use curly-brackets for this purpose.

Conditional Statement

Python if..else Statement Syntax:

```
if expression:  
    Statement  
elif expression:  
    Statement  
elif expression:  
    Statement  
....  
else:  
    Statement
```

Example

```
n = int(input("Enter a Number:"))  
if n%2==0:  
    print("Even")  
else:  
    print("Odd")
```

Result :

Enter a Number:10
Even

Enter a Number:11
Odd

Control statements

What are control Statements in Python?



Control statements in python are used to control the order of execution of the program based on the values and logic.

Control Statement

Python for Statement Syntax:

For in Loop

In Python, there is no C style for loop, i.e., for (i=0; i<n; i++). There is “for in” loop which is similar to for each loop in other languages.

Syntax:

```
for iterator_var in sequence:  
    statements(s)
```

Examples

```
x = [10,20,30,40,50,60]  
for i in x:  
    print(i)
```

Output:

10

20

30

40

50

60

Examples

```
x = [5,1,25,24,14,26,37,40]
even_list = []
odd_list = []
for i in x:
    if i%2==0:
        even_list.append(i)
    else:
        odd_list.append(i)
print(even_list)
print(odd_list)
```

output:

```
[24, 14, 26, 40]
[5, 1, 25, 37]
```

Control Statement

Python While Statement Syntax:

Syntax:

While Expression:
statements(s)

Examples

#to print the first 10 natural Numbers

```
i=1  
while i<=10:  
    print(i)  
    i+=1
```

**1
2
3
4
5
6
7
8
9
10**